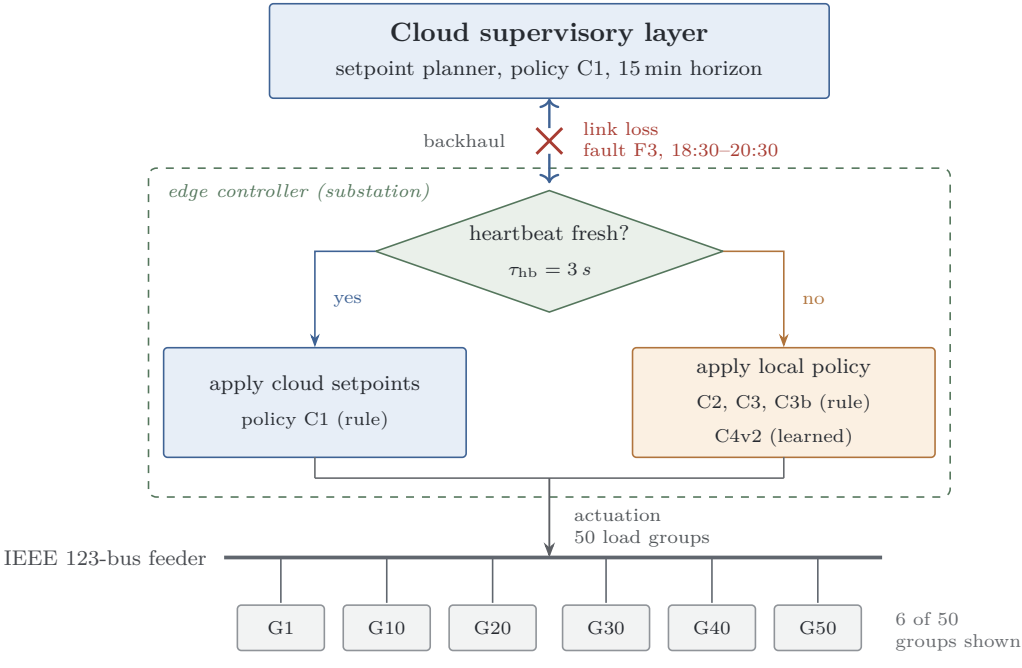

Cloud supervisory layer
setpoint planner, policy C1, 15 min horizon
backhaul
link loss
fault F3, 18:30–20:30
edge controller (substation)
heartbeat fresh?
$\tau_{hb} = 3\ s$
yes
no
apply cloud setpoints
policy C1 (rule)
apply local policy
C2, C3, C3b (rule)
C4v2 (learned)
actuation
50 load groups
IEEE 123-bus feeder
G1
G10
G20
G30
G40
G50
6 of 50
groups shown